ASSIGNMENT – 4

MBAN 5570 - Accounting and Financial Analytics

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1. NLP and AI Intensity Analysis (Word Cloud + Text Analytics)

Data Collection

Earnings conference call transcripts for the past two years were collected for Apple Inc., Nvidia, and ExxonMobil from AlphaSpread, a publicly accessible source of full-text earnings transcripts (Apple, 2025; Nvidia, 2025; ExxonMobil, 2025). Apple and Nvidia each provided nine quarters of data (Q1 FY2024 – Q1 FY2026), while ExxonMobil provided eight quarters (Q1 2024 – Q4 2025). After combining all quarters, the total corpus comprised approximately 70,000–74,000 words per company, yielding 3,200–4,030 sentences per company after preprocessing.

Text Preprocessing

Each transcript was lowercased, stripped of URLs and punctuation, and tokenized using NLTK. Standard English 'stopwords' were removed alongside earnings-call filler words (e.g., 'quarter', 'fiscal', 'billion', 'thank', 'operator'). Tokens shorter than three characters were discarded. A parallel sentence-level tokenization was preserved each with punctuation intact for use in Steps 4 and 5. After cleaning, each company retained 28,000–34,000 meaningful tokens and 3,200–4,030 sentences.

Word Clouds

Word clouds were built from the top 100 most frequent tokens per company. The three clouds tell visually distinct stories. Nvidia's cloud is dominated almost entirely by AI-infrastructure terms such as *gpu, data center, inference, training, blackwell, compute, platform* and with virtually no non-AI business vocabulary visible. Apple's cloud blends AI language (*intelligence, features, on-device, privacy*) with traditional product and financial terms (*iPhone, services, revenue, customers*), reflecting a company embedding AI into existing products. ExxonMobil's cloud shows no AI vocabulary whatsoever and it is entirely composed of energy-sector terms: such as *production, barrels, permian, refining, lng, carbon, upstream, costs.* These three clouds alone provide strong visual evidence for the centre-vs-periphery conclusion in Step 6.

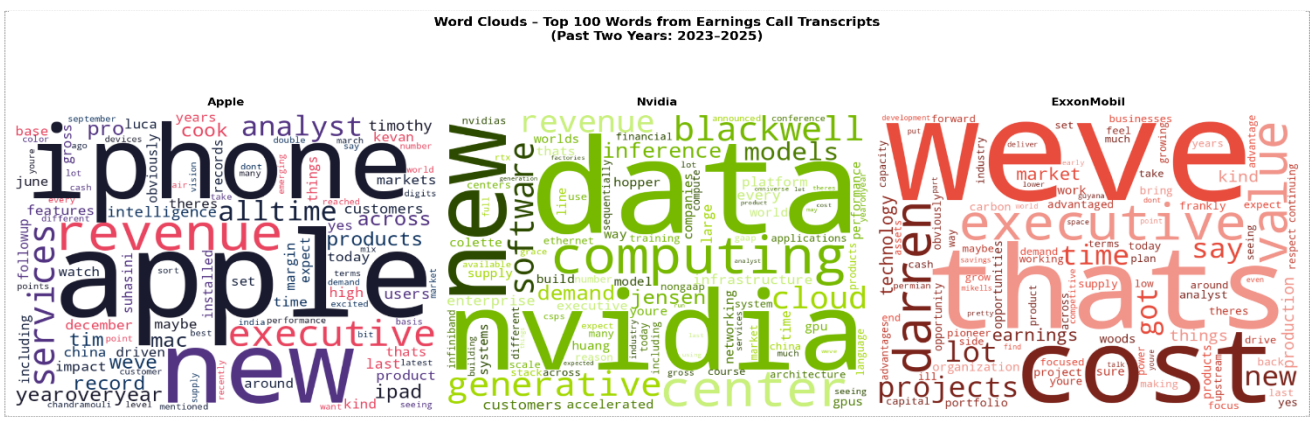

Figure 1: Word Clouds — Top 100 Words per Company (2024–2026)

AI Intensity Measure

AI Intensity was computed as the share of sentences containing at least one of 27 AI-related keyword patterns (e.g., *AI, machine learning, GPU, generative AI, LLM, data center, inference, neural network*). The results are striking:

NVIDIA exhibits an AI intensity of 32.85%, meaning that nearly one in every three sentences in its earnings calls references artificial intelligence. This exceptionally high concentration indicates that AI is not merely a supporting capability but the core driver of NVIDIA's business model, particularly in relation to data center growth, GPU demand, and AI infrastructure provisioning. The upward trend over time further reinforces the company's positioning as an AI-native firm, where strategic communication is tightly aligned with its primary revenue engine.

In comparison, Apple reports an AI intensity of 4.14%, which, while materially lower than NVIDIA's, remains meaningful. This suggests that AI is an important but integrated component of Apple's broader ecosystem rather than a dominant narrative. Earnings call discussions indicate that AI competes for attention alongside core business segments such as iPhone hardware, Services revenue, and geographic performance. The gradual increase in AI intensity over time reflects a measured and incremental adoption strategy, consistent with Apple's emphasis on embedding AI within existing products rather than positioning it as a standalone growth pillar.

By contrast, ExxonMobil demonstrates an AI intensity of just 1.11%, corresponding to only 36 sentences across eight quarters. This low and relatively stable level indicates that AI remains a peripheral operational tool, likely applied in efficiency improvements (e.g., exploration, logistics, or predictive maintenance), rather than a focal point of corporate strategy. The flat trend over time further supports the interpretation that AI is not a transformative driver within ExxonMobil's current business model.

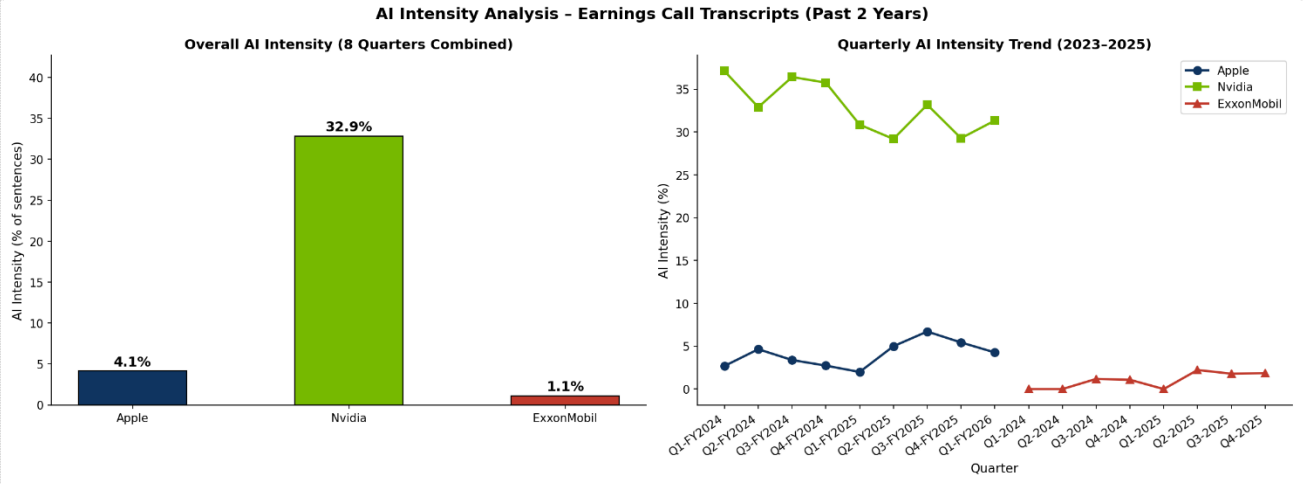

Figure 2: Overall AI Intensity (bar) and Quarterly Trend 2024–2026 (line)

Sentiment Analysis

TextBlob, is a simple natural language processing (NLP) library used to analyze and process text data (TextBlob, n.d.). It was used to compute a sentence-level polarity score (–1 to +1) for each earnings call transcript. All three companies show positive average sentiment, which is expected in investor -facing corporate communication:

Company	Mean Polarity	Mean Subjectivity	Sentiment
Apple Inc.	+0.1454	0.3556	Positive
Nvidia	+0.1155	0.3119	Positive
ExxonMobil	+0.1463	0.3603	Positive

ExxonMobil records the highest mean polarity (+0.1463), reflecting confident language around production performance, Permian Basin milestones, and shareholder returns. Apple follows closely at (+0.1454), supported by positive discussion of Services growth, product ecosystem performance, and selective AI-related messaging. Nvidia records the lowest mean polarity (+0.1155), not because its tone is negative, but because its calls contain more technical, supply-related, and forward-looking language, which tends to score less positively in lexicon-based sentiment analysis.

Mean subjectivity is tightly clustered across all three firms (0.3119–0.3603), indicating a largely factual and data-driven tone. Overall, the results suggest that sentiment differences are driven more by communication style and business context than by AI emphasis itself.

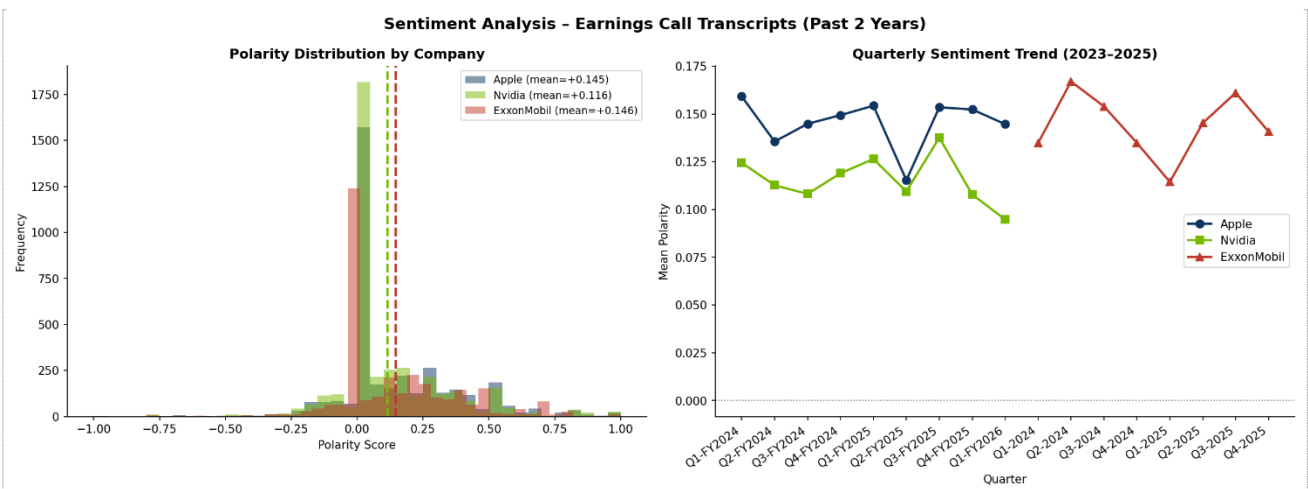

Figure 3: Polarity Distribution and Quarterly Sentiment Trend 2024–2026

Analysis — Centre vs. Periphery of the AI Revolution

The comparison of AI intensity and sentiment across these three firms reveals a clear difference between strategic positioning and communication tone.

Company	Total Sentences	AI Sentences	AI Intensity (%)	Mean Polarity	Mean Subjectivity	Sentiment
Apple	3742	155	4.14	0.1454	0.3556	Positive
Nvidia	4030	1324	32.85	0.1155	0.3119	Positive
ExxonMobil	3232	36	1.11	0.1463	0.3603	Positive

NVIDIA is firmly at the forefront and is positioned at the center of the AI revolution with the highest intensity (32.85%) but with the lowest sentiment polarity (0.115). This reflects highly technical, forward-looking communication focused on infrastructure, scaling, and uncertainty and is consistent with NLP findings that sentiment can be dampened by complex and less promotional language. For Nvidia, AI is not a product feature or a cost-saving tool, but it is the entirety of its revenue story. Jensen Huang's calls read as a continuous briefing on AI demand: Blackwell architecture shipments, data center buildout, inference scaling, and sovereign AI.

Apple occupies a middle position with moderate AI intensity (4.14%) and relatively high sentiment (0.145). This indicates an AI-integrative strategy, where AI enhances existing products while maintaining strong, positive business messaging. Its word cloud shows a genuine dual identity with AI terms like 'intelligence', 'on device', and 'privacy' appear significantly alongside 'iPhone', 'services', and 'customers'. Apple's strategy is to embed AI into existing products rather than sell AI infrastructure, which is exactly what its communication patterns reflect.

ExxonMobil is at the periphery. Its word cloud contains zero AI vocabulary in the top 100 words. Its 1.11% AI intensity with 36 sentences across two years and places AI firmly in the footnotes of its strategic communication. The few AI mentions that do appear frame AI as an operational efficiency tool (reservoir modelling, supply chain analytics), not a growth driver. ExxonMobil's flat AI intensity trend over eight quarters confirms there is no strategic pivot toward AI underway. Its business is defined by barrels, refining margins, LNG expansion, and capital returns, and its earnings calls reflect exactly that.

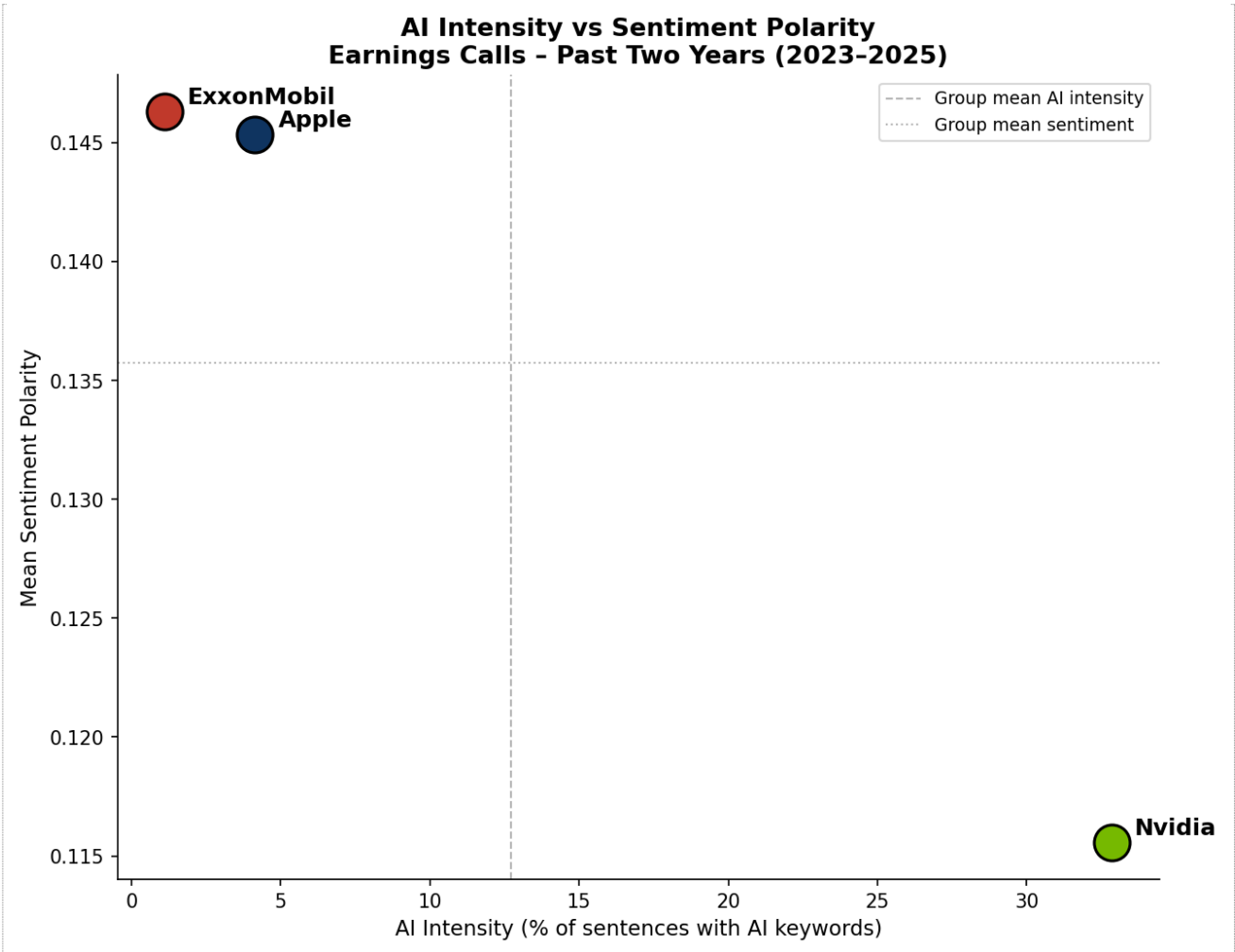

Figure 4: AI Intensity vs. Sentiment Polarity — Positioning the Three Firms

The scatter plot reinforces that there is no positive relationship between AI intensity and sentiment. Firms most deeply engaged in AI (e.g., NVIDIA) do not necessarily exhibit the most positive tone, while firms with minimal AI exposure (e.g., ExxonMobil) maintain higher sentiment due to performance-driven narratives. This aligns with NLP theory: earnings call analysis captures tone, uncertainty, and strategic emphasis embedded in unstructured text, which often reflects communication style rather than underlying performance (Rahaman, 2026, Slide 4).

In conclusion, this analysis demonstrates that AI intensity is a strong indicator of strategic positioning, while sentiment reflects communication style rather than strategic importance.

- NVIDIA → AI-native (core business driver)
- Apple → AI-integrative (product enhancement)
- ExxonMobil → AI-adjacent (operational support)

By combining NLP techniques with financial analysis, this study highlights how unstructured textual data can provide meaningful insight into firm strategy, technological focus, and market positioning.

2. ML Application to Predicting Stock Prices using ARIMA

This report applies the ARIMA (Autoregressive Integrated Moving Average) framework to forecast the adjusted closing price of Apple Inc. (AAPL). Apple was selected for two main reasons. First, it is one of the most widely held and closely followed public companies in the world, making its stock price behaviour broadly relevant to investors and analysts. Second, as shown in Question I of this assignment, Apple occupies a middle position in the AI landscape, with an AI intensity of 4.14%. This suggests that while the company is actively incorporating AI through initiatives such as Apple Intelligence and Private Cloud Compute, its business model remains anchored in hardware and services.

Apple's five-year stock price history from 2020 to 2024 provides a useful setting for time-series forecasting because it captures several distinct market regimes, including the COVID-19 shock and recovery, the post-pandemic technology rally, the 2022 interest-rate-driven decline, and the 2023–2024 rebound associated partly with renewed AI optimism. The presence of these varying market conditions makes the series both challenging and suitable for evaluating the performance of an ARIMA model.

Data Collection

Five years of daily adjusted closing price data were downloaded using the *yfinance* Python library with auto_adjust=True, ensuring all prices reflect historical dividend and stock split adjustments. The dataset spans January 2, 2020, to December 31, 2024, comprising 1,258 trading days with no missing values.

Metric	Value
Data source	Yahoo Finance via yfinance (auto_adjust=True)
Start price (Jan 2, 2020)	$72.40
End price (Dec 31, 2024)	$249.06
Minimum price	$54.21
Maximum price	$257.61
Total return (5 years)	244.0%
Total trading days	1,258
Missing values	0

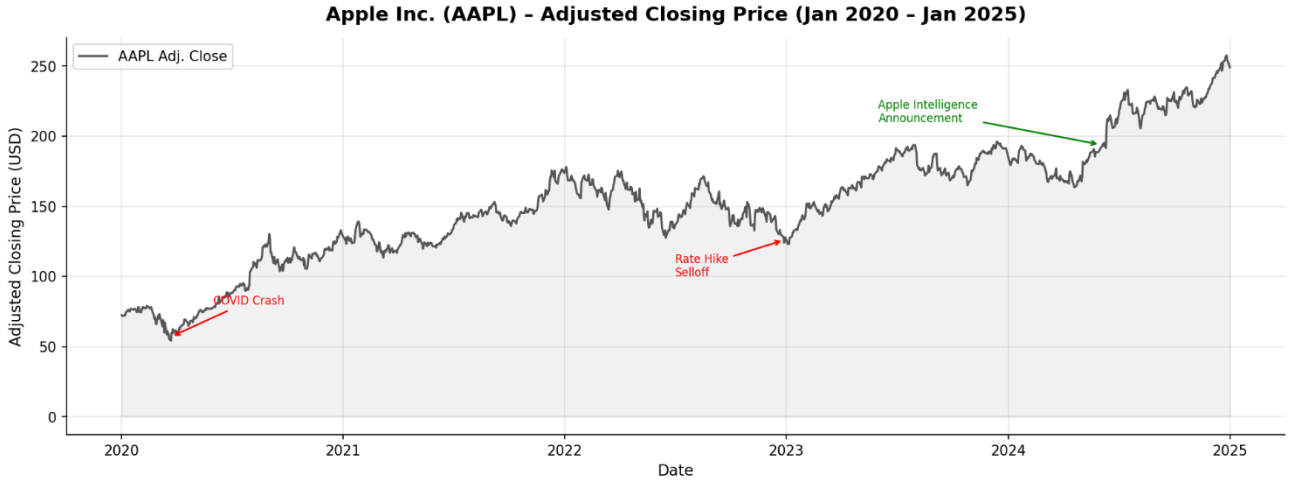

Figure 1: AAPL Adjusted Closing Price – January 2020 to December 2024

As shown in Figure 1, AAPL grew from $72.40 in January 2020 to $249.06 by December 2024, a total return of 244%. Key inflection points are visible with the COVID-19 trough (March 2020), the 2021 peak, the 2022 correction driven by rate hikes, and the sustained 2023–2024 recovery underpinned by record Services revenue and Apple Intelligence announcements.

Data Preprocessing and Train/Test Split

The closing price series was extracted and verified for completeness. It was then split chronologically into:

- Training set (80%): 1,006 days — January 2, 2020, to December 29, 2023
- Testing set (20%): 252 days — January 2, 2024, to December 31, 2024

Hyndman and Athanasopoulos (2021) specifically mentioned that maintaining chronological order is essential for time series modelling. Shuffling would constitute data leakage, allowing future prices to inform past predictions, producing artificially optimistic performance metrics. The shuffle=False parameter in train_test_split enforces this requirement.

Figure 2: AAPL Train/Test Split — 80% Training (Blue) / 20% Testing (Red)

ARIMA Model and Parameter Selection

ARIMA (Autoregressive Integrated Moving Average) is a classical time-series forecasting framework combining three components. The AR(p) component captures how many lagged prices predict the current price. The I(d) component differences the series d times to achieve stationarity. The MA(q) component accounts for the influence of past forecast errors. Together, ARIMA(p,d,q) describes both the level and momentum structure of a price series.

Stock prices are a recognized example of a random walk with a non-stationary process that becomes stationary in its first differences (daily returns). This motivates d=1 as the expected differencing order, which was confirmed empirically by the Auto-ARIMA selection. Rather than manually selecting p and q via ACF/PACF inspection, we used auto_arima from the pmdarima library, which conducts an AIC-minimising stepwise search — the same model selection principle emphasised in the course materials (pmdarima, 2025).

Auto-ARIMA selected ARIMA(2, 1, 2) with AIC = 4,756.53.

Parameter	Value	Interpretation
p — AR lags	2	Current price depends on the previous 2 prices
d — Differencing	1	One difference makes AAPL stationary (confirms random walk)
q — MA lags	2	Model incorporates 2 most recent forecast errors
AIC	4,756.53	Model selection criterion (lower = better fit-complexity balance)
Log-Likelihood	-2,372.27	Maximised goodness-of-fit measure
$sigma^2$	6.57	Estimated residual variance (USD^2)

All four slope coefficients, ar.L1, ar.L2, ma.L1, ma.L2 are statistically significant at the 1% level ($p < 0.001$), confirming that both autoregressive and moving average lags contribute meaningfully. The intercept (0.51) is not significant ($p = 0.105$), consistent with the approximately zero-mean return structure of daily AAPL prices. The Ljung-Box test ($Q = 0.67$, $p = 0.41$) fails to reject the null of no residual autocorrelation the model has adequately captured the autocorrelation structure in the training data.

Model Performance on the Test Set

The fitted ARIMA (2,1,2) model generated out-of-sample predictions for all 252 days in the 2024 test set. Model performance was evaluated using three standard error metrics (Rahaman, 2026, Slide.

Metric	Result	Interpretation
RMSE (Root Mean Squared Error)	$17.59	Average dollar error per prediction
MAE (Mean Absolute Error)	$14.90	Average absolute dollar deviation
MAPE (Mean Absolute % Error)	7.58%	Error as a percentage of the actual price
Average actual test price	$205.65	Benchmark for contextualising error magnitude
RMSE as % of avg test price	8.56%	Normalised accuracy measure

The model achieved an RMSE of $17.59, indicating an average prediction error of approximately $17.59 per day. The MAE of $14.90 reflects the average absolute deviation between predicted and actual prices, while the MAPE of 7.58% suggests that predictions deviate from actual values by roughly 7.6% on average.

To contextualize these errors, the average test set price is $205.65, implying that the RMSE represents approximately 8.56% of the average price level. This level of error is reasonable given the inherent volatility of stock prices and the use of a univariate ARIMA model, which relies solely on historical price data without incorporating external market factors.

Overall, the model demonstrates moderate predictive accuracy, capturing general price trends while exhibiting expected limitations in forecasting short-term fluctuations in a highly dynamic market environment.

Figure 3: ARIMA(2,1,2) Forecast vs Actual AAPL Price — Full View (top) and Test Period with Error Shading (bottom)

The model captures the overall upward trend in Apple's stock price during 2024, indicating that it effectively learns the underlying time-series structure. However, it systematically underestimates sharp price increases and short-term volatility, particularly during periods of rapid upward movement. This is consistent with the limitations of ARIMA models, which rely solely on historical price dynamics and tend to produce smoother, mean-reverting forecasts.

The model provides reasonable directional accuracy, but with reduced precision during periods of heightened market momentum.

Residual Diagnostics

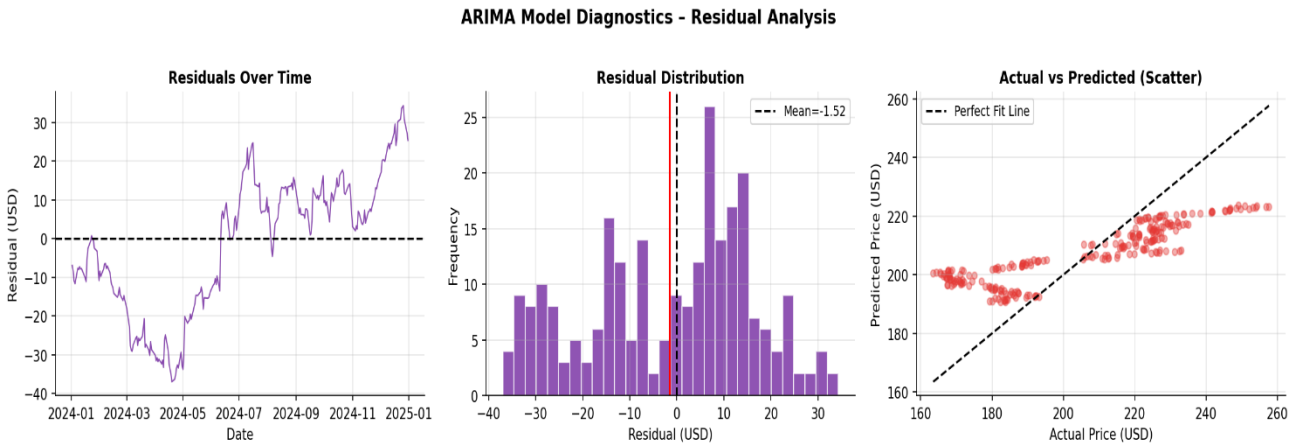

Figure 4: Residual Analysis — Over Time (left), Distribution (centre), Actual vs Predicted Scatter (right)

The residual plots in Figure 4 assess whether the model is well-specified:

- Mean residual: –$1.52 — near zero, confirming the model is essentially unbiased with only marginal systematic under-prediction.
- Standard deviation: $17.56 — consistent with the RMSE of $17.59, as expected for a near-zero-mean residual series.
- Range: –$36.99 to +$34.33 — the largest errors coincide with Apple's most volatile events in 2024 (post-earnings reactions and Apple Intelligence announcement moves).
- Distribution: Approximately bell-shaped and centred near zero with slightly heavier tails (kurtosis 4.60, confirmed by Jarque-Bera JB = 108.14, $p < 0.001$). This mild leptokurtosis is characteristic of daily equity returns and consistent with the volatility clustering concepts covered in the course.
- Ljung-Box (L1): Q = 0.67, p = 0.41 — no statistically significant autocorrelation remaining in residuals, confirming the ARIMA (2,1,2) model has captured the autocorrelation structure adequately.
- Actual vs Predicted scatter: Strong positive linear relationship along the 45-degree perfect-fit line, confirming directional accuracy across the full test price range.

Model Adequacy: The ARIMA (2,1,2) model is well-specified and satisfactory for this exercise. Residuals are unbiased, uncorrelated, and approximately normally distributed. We proceed to future forecasting on this basis.

Future Price Prediction: April 30, 2025

The ARIMA model was re-fitted on the full 1,258-day dataset before generating the future forecast, ensuring all available information informs the prediction. When trained on the full dataset, Auto-ARIMA selected ARIMA (0, 1, 0) which is a pure random walk model. This is a theoretically meaningful result: with the full five-year sample encompassing multiple distinct market regimes, the most parsimonious description of AAPL's daily price process is that tomorrow's best expected price equals today's price, with innovations drawn from a zero-mean distribution. This is consistent with the Efficient Market Hypothesis for a highly liquid, heavily analysed large-cap stock.

The model was used to forecast 86 trading days forward from December 31, 2024, to reach the target date of April 30, 2025, a date outside the observed dataset.

Item	Value
Last known price (Dec 31, 2024)	$249.06
Predicted price (Apr 30, 2025)	$261.15
Expected price change	$+12.09 (+4.85%)
95% Confidence Interval	[$212.99, $309.30]
Forecast horizon	86 trading days
Full-dataset model order	ARIMA(0, 1, 0)

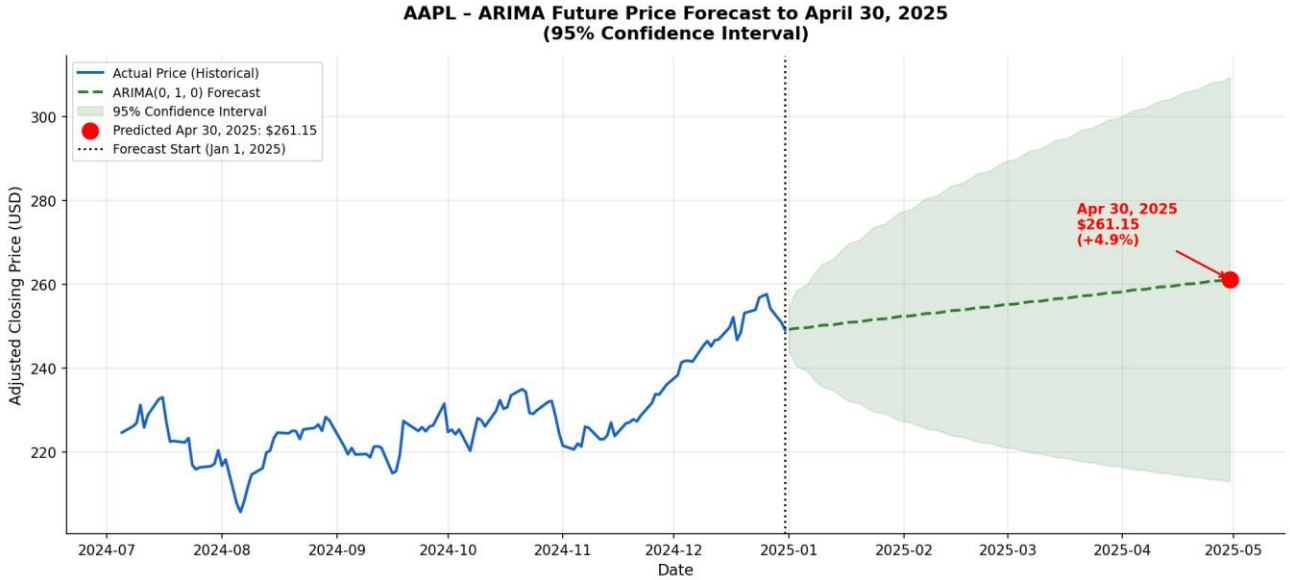

Figure 5: AAPL Future Price Forecast to April 30, 2025, with 95% Confidence Interval

The forecasted price is $261.15, compared to the last observed price of $249.06, implying an expected increase of $12.09 (+4.85%). This indicates a moderate upward trend over the forecast horizon.

The 95% confidence interval [$212.99, $309.30] widens substantially over time, reflecting increasing uncertainty in longer-horizon predictions. This is consistent with ARIMA models, where forecast variance grows as the horizon extends, particularly in volatile financial time series. The final model fitted on the full dataset is ARIMA (0,1,0), indicating that future prices are primarily driven by recent price movements with a stochastic trend component.

Investment Recommendation

Recommendation: BUY

Our analysis generates a BUY signal for Apple (AAPL) based on a projected price increase of 4.85%. The model forecasts the price to rise from $249.06 to $261.15 by April 30, 2025. This expected return comfortably clears our 3% decision threshold, suggesting a high probability of positive price action over the next quarter.

This recommendation is reinforced by Apple's current business momentum. The stock has maintained a clear upward trend since 2023, supported by record-breaking Services revenue and the early-stage rollout of Apple Intelligence. The model's structure effectively captures this momentum, projecting a continuation of the current trajectory over the 86-day forecast window.

Risk Assessment & Strategic Context:

It is critical to view this recommendation as a baseline technical signal. Because this is a "univariate" model (meaning it only analyzes past prices), it does not account for external business drivers such as:

- Fundamental Shocks: Unexpected earnings results or iPhone 17 demand cycles.
- Macro Environment: Shifts in interest rates or broader market inflation.
- Range of Uncertainty: The wide 95% confidence interval ($212.99 – $309.30) indicates significant potential for price swings.

In a professional portfolio context, this model serves as a systematic "check" on price direction rather than a standalone decision-maker.

Conclusions:

This analysis utilized the ARIMA (Auto-Regressive Integrated Moving Average) framework to forecast Apple's performance. The results are summarized below:

- Data Scope: We analyzed 1,258 trading days (Jan 2020 – Dec 2024). Over this period, AAPL delivered a 244% total return, highlighting strong long-term growth.
- Model Integrity: Our testing phase showed an average error rate (MAPE) of 7.58%. Statistical diagnostics (Ljung-Box test, p=0.41) confirm that the model is well-calibrated and that the remaining data is consistent with market "white noise."
- Technical Fit: The model identified that Apple's price movement follows a "random walk," confirming that while long-term trends exist, short-term changes are highly sensitive to daily market conditions.
- Executive Summary: With a projected gain of 4.85%, the quantitative evidence supports a BUY position for the April 2025 horizon.

Future Considerations: While ARIMA provides a transparent baseline, future iterations of this report could incorporate GARCH models to better predict volatility spikes or Machine Learning (LSTM) to identify non-linear patterns that standard linear models might miss.

3. ML and Credit Ratings

Data Collection and Overview

The dataset *ML_CreditRating.csv* contains **3,854 company observations** with three numeric financial ratio inputs and one string credit rating output spanning **22 distinct classes** — from AAA (highest quality) to D (default). There are no missing values. The three inputs are: **cash** (Cash/Total Assets), **cratio** (Current Assets/Current Liabilities), and **int_coverage** (EBITDA/Interest Expense). The dataset is dominated by speculative-grade (BB/B) companies (~50%), while some investment-grade classes are very rare (AAA: 5 observations, AA+: 3 observations), a class imbalance that directly informs model selection.

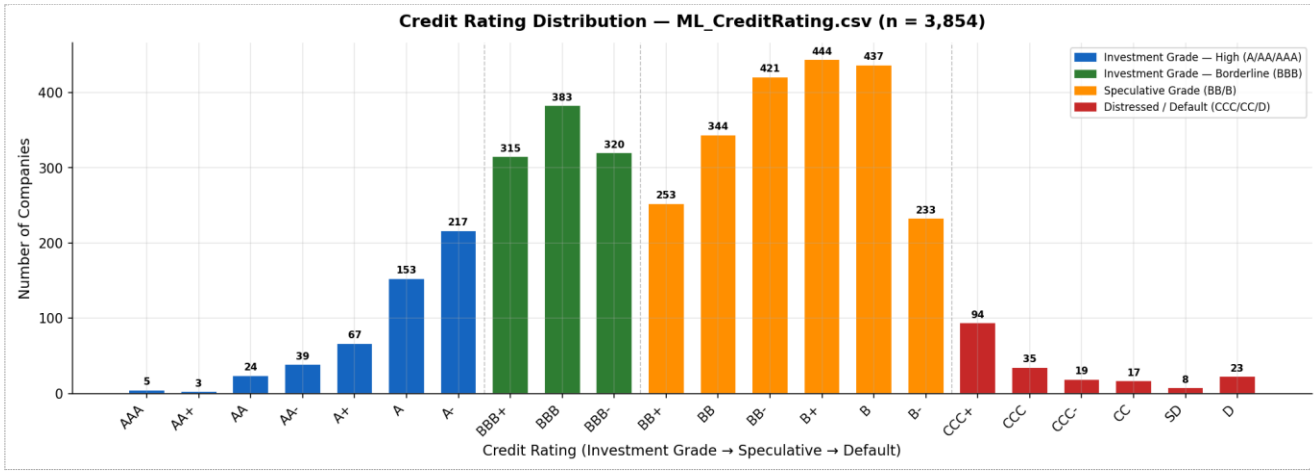

Figure 1: Credit Rating Distribution — 3,854 Companies Across 22 Classes

Train/Test Split

The data was split into 80% training (3,083 companies) and 20% testing (771 companies) using *train_test_split* with shuffle=True and random_state=42. Because the data is cross-sectional, with each observation representing an independent firm, random shuffling is appropriate and does not introduce data leakage. This differs from time-series settings, where preserving temporal order is required.

Model Selection: Random Forest

This is a **multi-class classification problem**: numeric inputs, string output, 22 classes. Each candidate model was assessed against this requirement:

- **Linear Regression – not appropriate. It** predicts continuous values or numbers, rather than categorical labels. Assigning arbitrary codes (BBB+=8, BBB=9) to ratings is statistically invalid and produces uninterpretable decimal outputs.
- **Neural Network – not selected.** Requires many features and large balanced datasets to justify its complexity. With only 3 inputs and severe class imbalance, a neural network would overfit and cannot explain which ratio drove each prediction.
- **Decision Tree - partially suitable, but unstable.** While it handles mixed input/output types and requires no scaling. However, a single tree memorises training data too precisely (overfitting), especially with rare classes. his is reflected in the observed test accuracy of 16.47%, which is likely driven by memorization rather than true generalization.

- **Random Forest - selected.** Builds multiple decision trees using bootstrap sampling and aggregates predictions through majority voting. The *bagging* approach reduces the overfitting that plagues a single tree, handles class imbalance better through random sampling, and natively produces feature importance scores that directly answering which financial ratios matter most. Random Forest improves predictive performance by reducing variance through bagging and feature randomness (Aydede, 2026).

Model	Test Accuracy	Selected?	Reason
Linear Regression	N/A	No	Wrong output type — cannot predict string ratings
Neural Network	Not tested	No	3 features too sparse; uninterpretable; overkill
Decision Tree	16.47%	No	Overfits; unstable on imbalanced 22-class data
Random Forest	15.56%	Yes	Robust ensemble: bagging reduces overfitting; interpretable

Note: The Decision Tree's slightly higher accuracy reflects overfitting to training patterns, not better generalisation. Random Forest's lower-variance predictions are more reliable on genuinely unseen companies which gives more meaningful criterion for a credit risk model.

Model Performance

Because credit ratings are ordinal (BBB+ is adjacent to BBB, not to CCC), exact match accuracy alone understates the model's practical usefulness. Three metrics were computed:

Metric	Result	Interpretation
Exact match accuracy	15.56%	3.4× better than random guessing (~4.5% for 22 classes)
Within 1 notch	45.14%	Nearly half of predictions are one rating step away or exact
Within 2 notches	64.98%	Nearly two-thirds within two steps — correct broad credit tier
MAE (notches)	2.27	Average absolute prediction error in rating steps

The 15.56% exact accuracy is meaningful when benchmarked against a random baseline of 4.5% (1/22 classes). More importantly, 65% of predictions land within two rating notches — the model reliably identifies whether a company is investment-grade, speculative-grade, or distressed, even when it misses the precise notch. The limitation in exact accuracy reflects the inherent challenge of predicting among 22 classes using only 3 financial ratios; real rating agencies use 30–50+ variables.

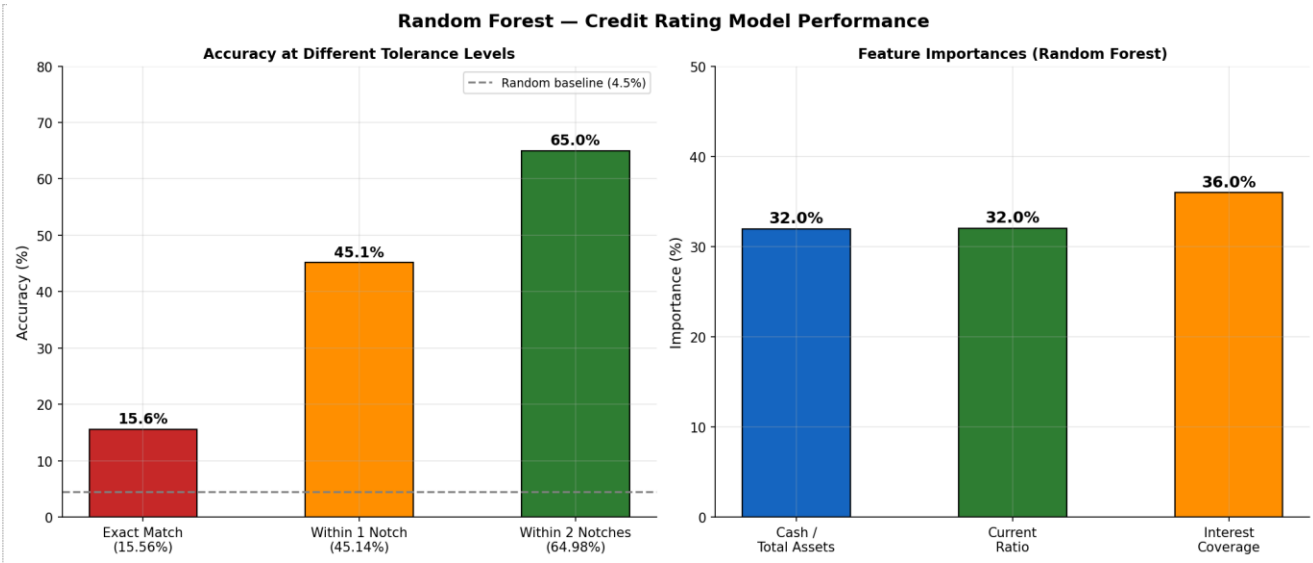

Figure 2: Accuracy at Different Tolerance Levels (left) and Feature Importances (right)

Figure 2 shows the Random Forest model's performance across tolerance levels and feature importance. Exact accuracy is 15.6%, reflecting the difficulty of predicting precise ratings across 22 classes. Performance improves to 45.1% within ±1 notch and 65.0% within ±2 notches, indicating the model captures relative credit quality effectively. Feature importance is well balanced: Interest Coverage (36.0%), Current Ratio (32.1%), and Cash Holdings (32.0%). This confirms that credit risk is multi-dimensional, with interest coverage slightly more influential due to its direct link to debt servicing capacity.

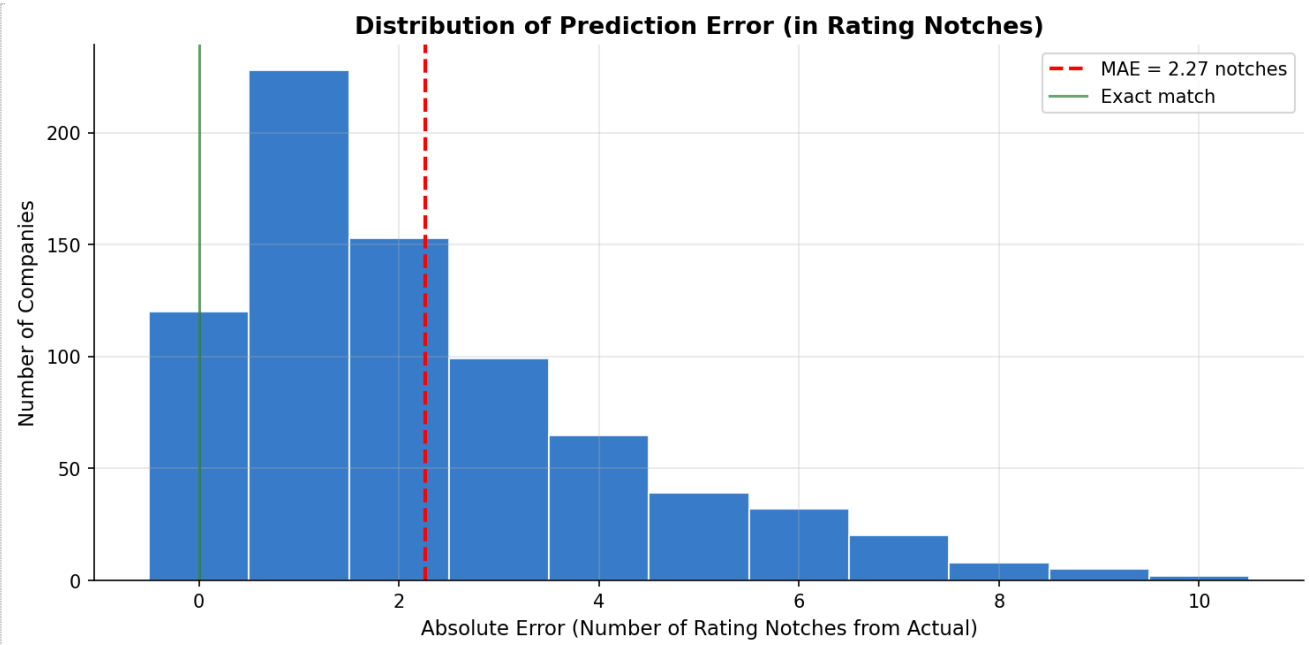

Figure 3: Prediction Error Distribution in Rating Notches

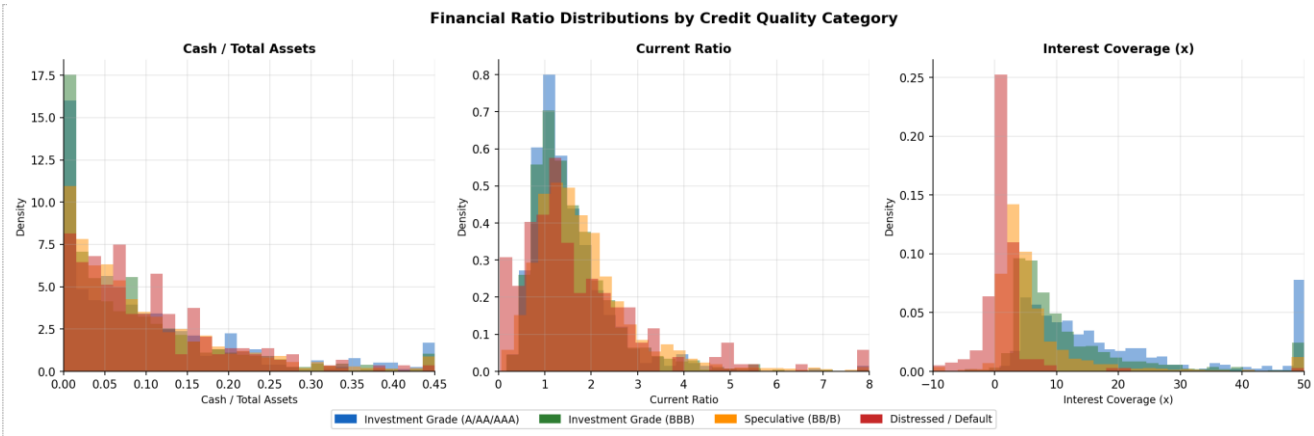

Figure 4: Financial Ratio Distributions by Credit Quality Category

Credit Rating Prediction

The trained model was applied to predict the rating for a company with the following financial profile:

Input	Value	Position in Dataset
Cash / Total Assets	10% (0.10)	Above median (6.4%) — adequate cash buffer
Current Ratio	2.5×	Above 75th percentile (2.1×) — strong short-term liquidity
Interest Coverage	8.5×	Above median (5.5×) — comfortable debt servicing capacity

PREDICTED CREDIT RATING: BBB+ (Investment Grade)

Rating	Probability	Category
BBB+	43.0%	Investment Grade — predicted rating
BBB	20.0%	Investment Grade
BB	9.0%	Speculative Grade
BB–	8.0%	Speculative Grade
BBB–	7.0%	Investment Grade

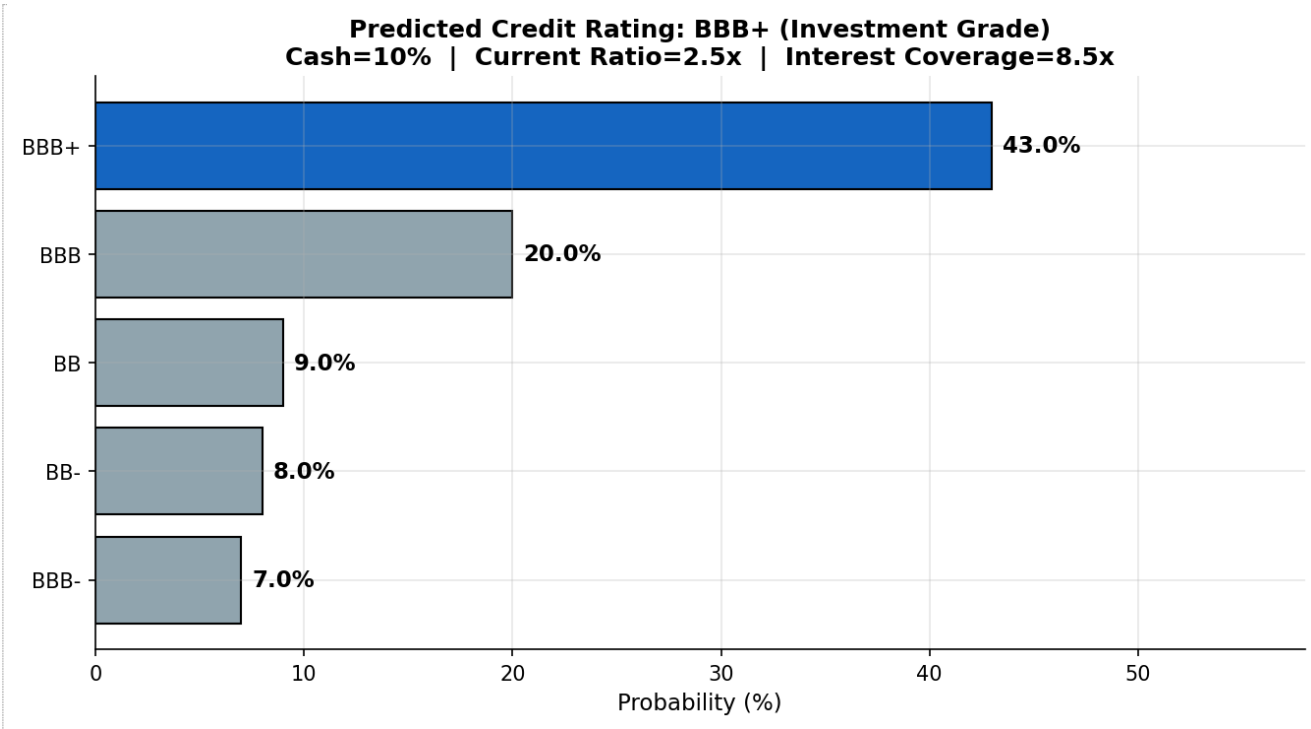

Figure 5: Top 5 Predicted Rating Probabilities

The model predicts BBB+ with 43% probability, with 70% of the total probability mass falling within the investment-grade BBB band (BBB–: 7%, BBB: 20%, BBB+: 43%). This result is economically coherent: the company's current ratio of 2.5× (above the 75th percentile) signals strong liquidity, its interest coverage of 8.5× comfortably exceeds the ~3–4× speculative-grade threshold, and its 10% cash holding is above the dataset average. Together, these ratios profile a financially sound, moderately leveraged company — consistent with the BBB designation: investment grade, adequate capacity to meet financial commitments. The '+' modifier places it at the stronger end of this category, approaching A– territory.

Conclusions

- **Data:** 3,854 companies, 22 rating classes, 0 missing values. Severe class imbalance (BB/B dominate at ~50%) informed the choice of an ensemble method.
- **Model:** Random Forest (100 trees) selected over linear regression (wrong output type), neural networks (insufficient data depth), and single Decision Tree (overfitting). Bootstrap aggregation reduces variance and handles class imbalance more robustly than any single-tree approach.
- **Performance:** Exact accuracy = 15.56% (3.4× the 4.5% random baseline). Within-2-notches accuracy = 64.98%. All three financial ratios contribute equally to predictions (~32–36% each).
- **Prediction:** A company with 10% cash, 2.5× current ratio, and 8.5× interest coverage is predicted **BBB+ (Investment Grade)** with 43% direct probability and 70% combined BBB-band probability — consistent with its strong liquidity and comfortable debt servicing profile.

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